RACE Regular Analysis Through Continuous Exercise

Gravitation : RACE-01

1. If R is the radius of the earth and the acceleration due to gravity on the surface of earth is $g = \pi^2\ m/s^2$, then the length of the second's pendulum at a height $h = 2R$ from the surface of earth will be:

(A) $\frac{2}{9}m$ (B) $\frac{1}{9}m$ (C) $\frac{4}{9}m$ (D) $\frac{8}{9}m$

PGT106

2. At what distance above and below the surface of the earth a body will have same weight, (take radius of earth as R.)

(A) $\sqrt{5}R - R$ (B) $\frac{\sqrt{3}R - R}{2}$ (C) $\frac{R}{2}$ (D) $\frac{\sqrt{5}R - R}{2}$

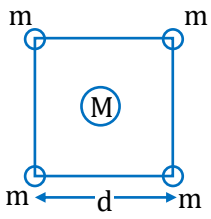
PGT107

3. If the gravitational field in the space is given as $\left(-\frac{K}{r^2}\right)$. Taking the reference point to be at $r = 2\ cm$ with gravitational potential $V = 10\ J/kg$. Find the gravitational potential at $r = 3\ cm$ in SI unit: (Given, that $K = 6\ J\ cm/kg$)

(A) 9 (B) 11 (C) 12 (D) 10

PGT108

4. Four spheres each of mass m form a square of side d (as shown in figure). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is :

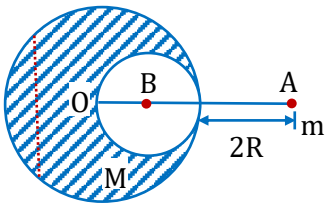


(A) $-\frac{Gm}{d}\left[(4+\sqrt{2})m + 4\sqrt{2}M\right]$ (B) $-\frac{Gm}{d}\left[(4+\sqrt{2})M + 4\sqrt{2}m\right]$

(C) $-\frac{Gm}{d}\left[3m^2 + 4\sqrt{2}M\right]$ (D) $-\frac{Gm}{d}\left[6m^2 + 4\sqrt{2}M\right]$

PGT109

5. A solid sphere of radius R gravitationally attracts a particle placed at $3R$ form its centre with a force F_1. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of $F_1 : F_2$ is :

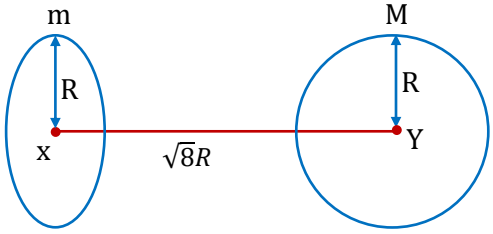


(A) 25 : 36 (B) 36 : 25 (C) 50 : 41 (D) 41 : 50

PGT110

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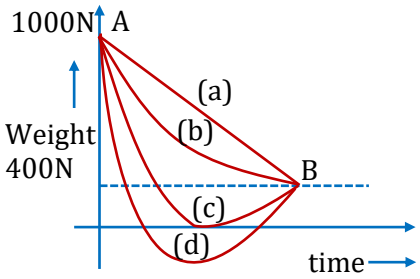
6. Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}R$ is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.



(A) $\frac{\sqrt{8}R}{9}.\frac{GmM}{R}$ (B) $\frac{2\sqrt{2}}{3}.\frac{GMm}{R^2}$ (C) $\frac{1}{3\sqrt{8}}.\frac{GMm}{R^2}$ (D) $\frac{\sqrt{8}}{27}.\frac{GmM}{R^2}$

PGT111

7. A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other objects in sky and take acceleration due to gravity on the surface of the Earth and Mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time.



(A) (c) (B) (a)
(C) (d) (D) (b)

PGT112

8. A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____S.

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(A) $\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$ (B) $\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

(C) $\frac{1}{3}\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$ (D) $\frac{1}{3}\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

PGT113

9. The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r)=\rho_0\left(1-\frac{r^2}{R^2}\right)$. Then the gravitational field is maximum at:

(A) $r=\frac{1}{\sqrt{3}}R$ (B) $r=\sqrt{\frac{5}{9}}R$ (C) $r=\sqrt{\frac{3}{4}}R$ (D) $r = R$

PGT114

10. The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω. An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is : ($h << R$, where R is the radius of the earth)

(A) $\frac{R^2\omega^2}{8g}$ (B) $\frac{R^2\omega^2}{4g}$ (C) $\frac{R^2\omega^2}{g}$ (D) $\frac{R^2\omega^2}{2g}$

PGT115

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	A	C	B	A	D	B	D

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RACE Regular Analysis Through Continuous Exercise

Gravitation : RACE-02

1. The escape velocities of two planets A and B are in the ratio 1 : 2. If the ratio of their radii respectively is 1 : 3, then the ratio of acceleration due to gravity of planet A to the acceleration of gravity of planet B will be:

(A) $\frac{4}{3}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

PGT116

2. If earth has a mass nine times and radius twice to the of a planet P. Then $\frac{v_e}{3}\sqrt{x}\ ms^{-1}$ will be the minimum velocity required by a rocket to pull out of gravitational force of P, where v_e is escape velocity on earth. The value of x is

(A) 2 (B) 3 (C) 18 (D) 1

PGT117

3. Two satellites A and B having masses in the ratio 4 : 3 are revolving in circular orbits of radii $3r$ and $4r$ respectively around the earth. The ratio of total mechanical energy of A to B is :

(A) 9 : 16 (B) 16 : 9 (C) 1 : 1 (D) 4 : 3

PGT118

4. A body of mass m is projected with velocity λv_e in vertically upward direction from the surface of the earth into space. It is given that v_e is escape velocity and $\lambda < 1$. If air resistance is considered to the negligible, then the maximum height from the centre of earth, to which the body can go, will be (R : radius of earth)

(A) $\frac{R}{1+\lambda^2}$ (B) $\frac{R}{1-\lambda^2}$ (C) $\frac{R}{1-\lambda}$ (D) $\frac{\lambda^2 R}{1-\lambda^2}$

PGT119

5. If one wants to remove all the mass of the earth to infinity in order to break it up completely. The amount of energy that needs to be supplied will be $\frac{x}{5}\frac{GM^2}{R}$ where x is _____.

(Round off to the Nearest Integer) (M is the mass of earth, R is the radius of earth, G is the gravitational constant)

PGT120

6. A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R (R = Radius of the earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the Earth)

(A) $\frac{m}{20}\left(u-\sqrt{\frac{2GM}{3R}}\right)^2$ (B) $5m\left(u^2-\frac{119}{200}\frac{GM}{R}\right)$ (C) $\frac{3m}{8}\left(u+\sqrt{\frac{5GM}{6R}}\right)^2$ (D) $\frac{m}{20}\left(u^2+\frac{113}{200}\frac{GM}{R}\right)$

PGT121

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7. An asteroid is moving directly towards the centre of the earth. When at a distance of $10R$ (R is the radius of the earth) from the earths centre, it has a speed of 12 km/s. Neglecting the effect of Earth's atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is 11.2 km/s)?
Give your answer to the nearest integer in kilometer/s _____.

PGT122

8. The mass density of a spherical galaxy varies as $\frac{K}{r}$ over a large distance 'r' from its centre. In that region, a small star is in a circular orbit of radius R. Then the period of revolution, T depends on R as:

(A) $T \propto R$ (B) $T^2 \propto \frac{1}{R^3}$ (C) $T^2 \propto R$ (D) $T^2 \propto R^3$

PGT123

9. A satellite is in an elliptical orbit around a planet P. It is observed that the velocity of the satellite when it is farthest from the planet is 6 times less than that when it is closest to the planet. The ratio of distances between the satellite and the planet at closest and farthest points is :

(A) 1 : 6 (B) 3 : 4 (C) 1 : 3 (D) 1 : 2

PGT124

10. The energy required to take a satellite to a height 'h' above Earth surface (radius of Earth = 6.4×10^3 km) is E_1 and kinetic energy required for the satellite to be in a circular orbit at this height is E_2. The value of h for which E_1 and E_2 are equal, is:

(A) 1.28×10^4 km (B) 6.4×10^3 km (C) 3.2×10^3 km (D) 1.6×10^3 km

PGT125

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	B	B	3	B	16	C	A	C

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